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Assessing Scaling Risks in UAE Carbonate Reservoirs Using Reactive Transport Modelling

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Abstract

This study aims to investigate the impact of in situ reactions on scaling risk due to various injection water types in a giant offshore carbonate reservoir in Abu Dhabi. The goal is to understand how different water compositions, temperatures, and injection strategies affect the formation of calcium sulfate ($CaSO_4$) and strontium sulfate ($SrSO_4$) scales, and to provide recommendations for managing and mitigating these scaling risks to reduce operational costs.

The study employs reactive transport modelling tools (CMG GEM) to simulate the impact of various factors such as injection water composition, temperature, rates, and the location of injection wells relative to production wells. The model also considers features like the oil-water contact and gas-oil contact. The output includes ion concentration profiles, water composition profiles, and water production rates at production wells impacted by seawater breakthrough. These profiles help identify the evolution of scaling risks in the production wells.

The GEM model predicts no $SrSO_4$ scaling risk for any of the producers, but there is a $CaSO_4$ scaling risk for all three producers that cut seawater. The high temperature of the formation (104°C) plays a critical role in reducing sulfate scaling risk due to the natural precipitation of $CaSO_4$ deep within the reservoir. The study concludes that the reservoir itself acts as a natural desulfation plant, depleting the brines of scaling ions and thereby protecting the wells. The findings suggest that installing a Sulfate Reduction Plant (SRP) is not necessary. Matrix transport has been considered in this study. The effect of the presence of fractures acting as a "short circuits" high conductivity pathways is discussed in the paper.

The novelty of this study lies in its use of reactive transport modelling to account for in situ geochemical reactions, providing a more accurate assessment of scaling risks. The study's findings challenge the conventional need for SRPs by demonstrating that the reservoir itself can act as a natural desulfation plant. This approach not only reduces operational costs but also offers new insights into scale management strategies for similar fields.

Introduction

Umm Shaif field is located 140 km offshore North-West of Abu Dhabi city (Figure 1). The operations are centered on producing oil and gas from these fields and transferring the crude through a sophisticated pipeline network to Das Island for processing, storing and exporting (Figure 2). Water Injection for the maintenance of the reservoir pressure has been applied at both the Umm Shaif and the Lower Zakum fields since 1976.

The injection of water is used for pressure support and to sweep oil towards production wells. The presence of sulfate (SO_4^{2-}) anions in seawater and cations such as calcium (Ca^{2+}) and strontium (Sr^{2+}) in the formation water may lead to the deposition of sulfate scales such as calcium sulfate ($CaSO_4$) and strontium sulfate ($SrSO_4$). When the mixing occurs in or very close to the production well, this may cause significant loss of production due to loss of permeability or due to restrictions within the wellbore and production system. Control of this deposition may be achieved by altering the composition of the injection water to reduce the SO4 concentration, such as by dilution of seawater with another lower sulfate brine, use of low sulfate seawater from a sulfate reduction plant (SRP), or by injecting a low sulfate or sulfate free aquifer water.

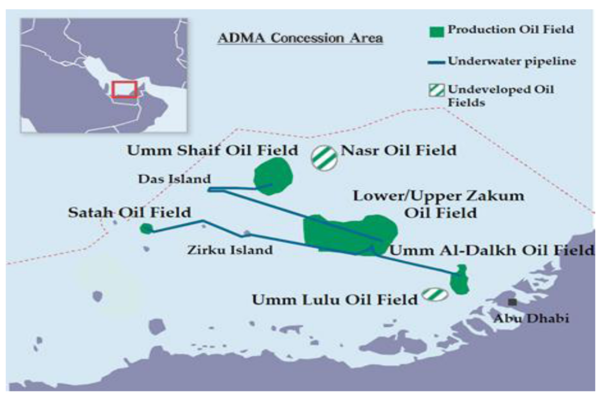

Figure 1—Umm Shaif field location

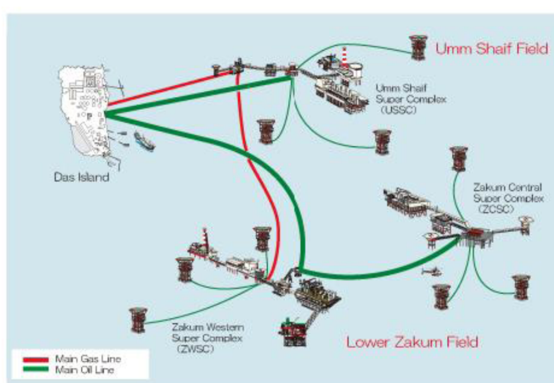

Figure 2—View of main gas line and main oil line for Umm shaif field

If SRP is used, relaxation of the requirements to minimize the SO_4^{2-} concentration, say over time, can significantly reduce operational costs. It is also possible that reactions in the reservoir may deplete $SO4^{2-}$ from the injection brine stream, helping to mitigate the risk. The fact that the Arab C/D field (Umm Shaif reservoir) is at 104°C may prove critical in this regard, since $CaSO_4$ is relatively soluble at downhole water injection temperatures (<50°C), but above 90°C the solubility decreases significantly. This means that as the injection water heats up, the presence of Ca^{2+} in the formation water (and in the rock) can lead to a natural sulfate reduction mechanism due to precipitation of $CaSO_4$ deep within the reservoir, reducing the sulfate scaling risk in the production system. This effect has been observed in the Ekofisk field, a chalk reservoir at 130°C in the North Sea. In such systems, the presence of bicarbonate (HCO_3^-) in the brines, and the reactions with the carbonate rock can have a significant impact on the abundance of Ca^{2+} that drives the $CaSO_4$ reaction. In the Umm Shaif field all these factors may play a role in determining the sulfate scaling risk at the production wells, how this risk evolves over the lifetime of the field, and under the injection of different water compositions. They also determine when chemical scale inhibitor treatments may be required, how they should be deployed and how frequently they should be repeated.

This study uses reactive transport modeling tools (CMG GEM) to examine the effects of injection water composition, temperature, rates, the location of injection wells in relation to production wells, and characteristics such as the oil-water contact and gas-oil contact, to evaluate the impact on calcium sulfate and strontium sulfate scaling risks in the Arab C/D field, with resulting recommendations for options for managing and mitigating the scaling risk and for reducing operating costs.

The output of these calculations are ion concentration profiles along with water composition profiles and water production rates at the production wells impacted by seawater breakthrough, which can then be used to identify the evolution of the scale risk in the production wells. Since the concentration of strontium in the produced brine declines due to dilution of the originally strontium-rich formation water with strontium free injection water as the injection water fraction in the produced brine increases, the concentration of sulfate that may be tolerated in the produced brine increases over time. The precipitation deep within the reservoir is known not to affect porosity or permeability to any significant extent - the damage only occurs in the near wellbore zone and the well itself.

CMG GEM modelling

A CMG GEM compositional reactive transport model input dataset was developed based on an ECLIPSE flow model input data set supplied by ADNOC, adapted to include PVT properties based on an Equation of State (Peng Robinson), also provided by ADNOC. The GEM model couples aqueous component transport, aqueous speciation reactions, mineral precipitation and dissolution reactions (using the Pitzer model) as well as heat transport. It is very important to consider heat transport as the anhydrite reaction is temperature dependent (low solubility at higher temperature, shown in Figure 3).

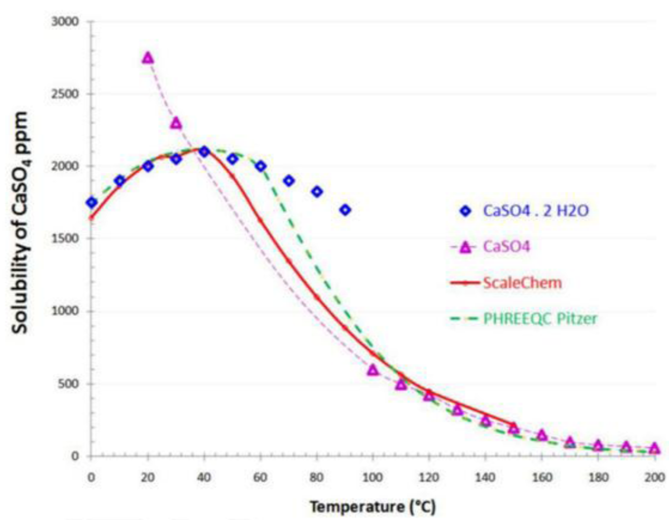

Figure 3—Solubility of $CaSO_4$ (gypsum at T < 90°C and anhydrite at T > 90°C) in deionized water as a function of temperature: solid diamond and triangle symbols represent experimental data, the solid red line is calculated using the ScalChem thermodynamic model from OLI Systems, and the dotted green line using PHREEQC with the Pitzer database (Lopez-Salinas et al., 2011)

The GEM model includes two initial minerals in the reservoir: dolomite ($CaMg(CO_3)_2$) and calcite ($CaCO_3$). Dolomite is to represent dolomite or similar calcium magnesium carbonate minerals that may appear in the reservoir; calcite is the main source for calcium in the calcium-rich formation water.

The mineralogy and reactions considered in the model are described below; where initial minerals are dolomite and calcite (Eqs 1 and 2) and Eqs 3 and 4 describe the precipitations equations for anhydrite ($CaSO_4$) and celestite ($SrSO_4$).

$$\text{Dolomite: } Ca + Mg + 2CO_3 \leftrightarrow CaMg(CO_3)_2 \quad \text{Eq. 1}$$

$$\text{Calcite: } Ca + CO_3 \leftrightarrow CaCO_3 \quad \text{Eq. 2}$$

$$\text{Anhydrite: } Ca + SO_4 \leftrightarrow CaSO_4 \quad \text{Eq. 3}$$

$$\text{Celestite: } Sr + SO_4 \leftrightarrow SrSO_4 \quad \text{Eq. 4}$$

Equation 3 describes the precipitation at reservoir temperature, not at injection temperature. Equations 1 to 4 show that two ions are common in these equations: calcium and sulfate.

Based on the chemical reactions identified above, the common ions in this simulation study are calcium (which is involved in three different reactions), and sulfate (which is involved in two reactions), with the anhydrite reaction involving both these ions. The speciation reactions of H_2S and CO_2 are also included in the compositional model as a consideration to account for any possible pH change, which will impact the carbonate mineral reactions.

Model initialization

Table 1 shows the ionic composition of the formation water (original and equilibrated with calcite and dolomite at reservoir conditions) and the injection water. It can be seen from Table 1 that the equilibrated formation water has different ion concentrations compared to the original formation water compositions provided. There is an increase in Ca^{2+} along with decreases in Mg^{2+}, $SO4^{2-}$ and HCO_3^- concentrations. There may be various reasons for these variations: contamination of samples, preservation of samples, method used in reconstructing the brine composition to be equilibrated at reservoir conditions taking account of the mole fraction concentrations of CO2 (3.758%) and H2S (2.092%) in the hydrocarbon phase, model databases, etc. Therefore, the relatively high concentration of $SO4^{2-}$ in the supplied formation water composition (235 mg/L) would lead to instantaneous anhydrite precipitation in the model, even before the simulation of water injection began. The consumption of Ca^{2+} would then lead to calcite dissolution, which would in turn increase the CO_3^{2-} concentration, resulting in dolomite precipitation, which is what leads to the decrease of Mg^{2+} and HCO_3^- in the equilibrated brine. Nonetheless, once the equilibration process has been completed, the change in brine composition due to the injection of a non-native brine will disrupt the equilibrium and lead to precipitation and dissolution reactions, with the consequent impact on the produced brine compositions and thus the scaling risk.

Literature review has shown that some stripping of sulfate occurs in the reservoir (Al Kalbani et al., 2020, Al Riyami et al., 2008, Ishkov and Mackay, 2018, Vazquez and al., 2013; Wang et al., 2024)). To evaluate the level of sulfate at the vicinity of the wellbore area of the production well, a geochemical reservoir model accounting for dispersion, degree of heterogeneity, size of the reservoir, mineralogy, petrophysics, degree of heterogeneity, presence of barriers, position of injectors and producers and in situ reaction rate is required (Baraka-Lokmane et al., 2023).

Table 1—Water composition for formation water (original and equilibrated with calcite and dolomite at reservoir conditions) and injection water

Ions	Formation water (mg/L)	Sea water (mg/L)	Formation water Equilibrated (mg/L)
Na	73900	12066	73900
K	4690	5035	4690
Mg	2590	1482	489
Ca	21800	8192	25164
Ba	39	2	39
Sr	700	164	700
Fe	03	0	0.3
Cl	165770	22505	165770
Br	0	0	0
SO_4	235	3047	55
HCO_3	190	168	44
TDS	269879	40609	270816

Figure 4 shows the locations of the three producers (P1, P6 and P7) that undergo seawater breakthrough and main water injector (WI4), along with other wells in the study area. Injector WI4 is located above the oil water contact and producer P7 is the one located closest to the oil water contact. The other producer and injector well locations are included for reference but are less significant in this study as there is no sulfate scaling risk in these wells.

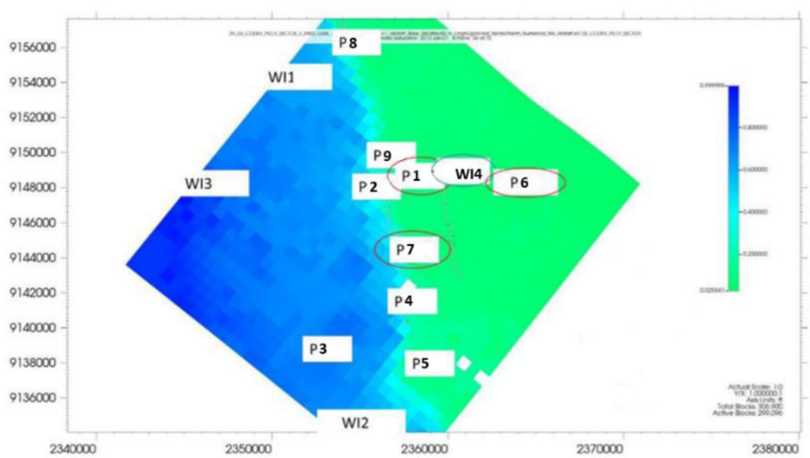

Figure 4—Locations of producers that experience seawater breakthrough and main water injector superimposed on map showing initial water saturation in GEM model, layer 36 (out of a total of 75 layers). (Locations are identically the same in the ECLIPSE model).

Modelling results

Comparison between developed GEM model and original supplied ECLIPSE model

Figures 5 to 7 show the calculated oil, gas, water and liquid rates for three producers (P1, P6 and P7) in the study area, comparing the values calculated using the developed GEM model with the values calculated by the original ECLIPSE model. Both models start (i.e. time = 0 on the axis) on 1st January 2012.

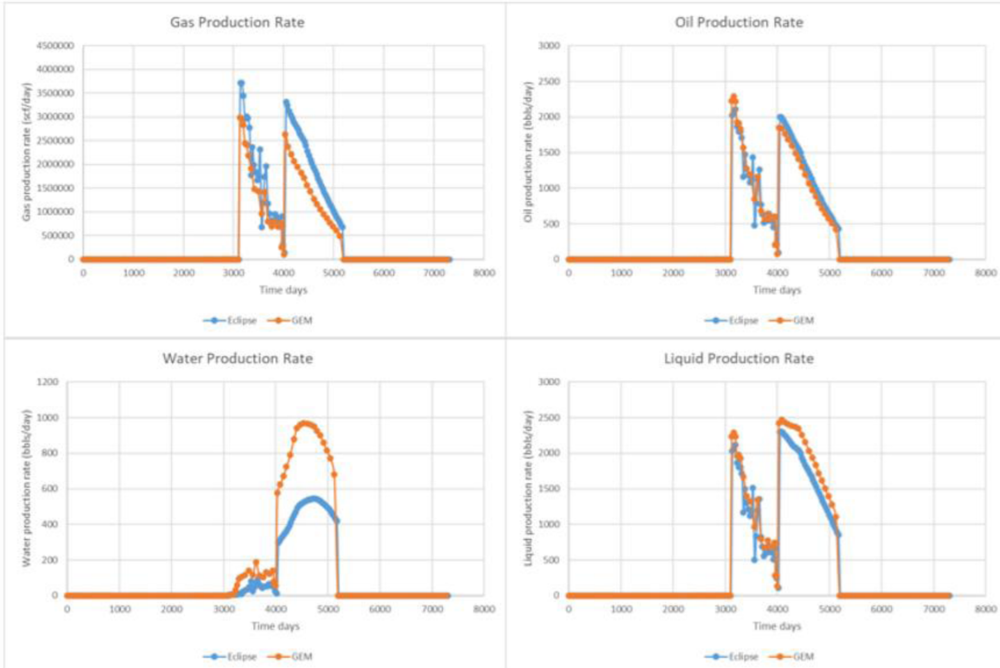

Figure 5—ECLIPSE and GEM Comparison – Well P1

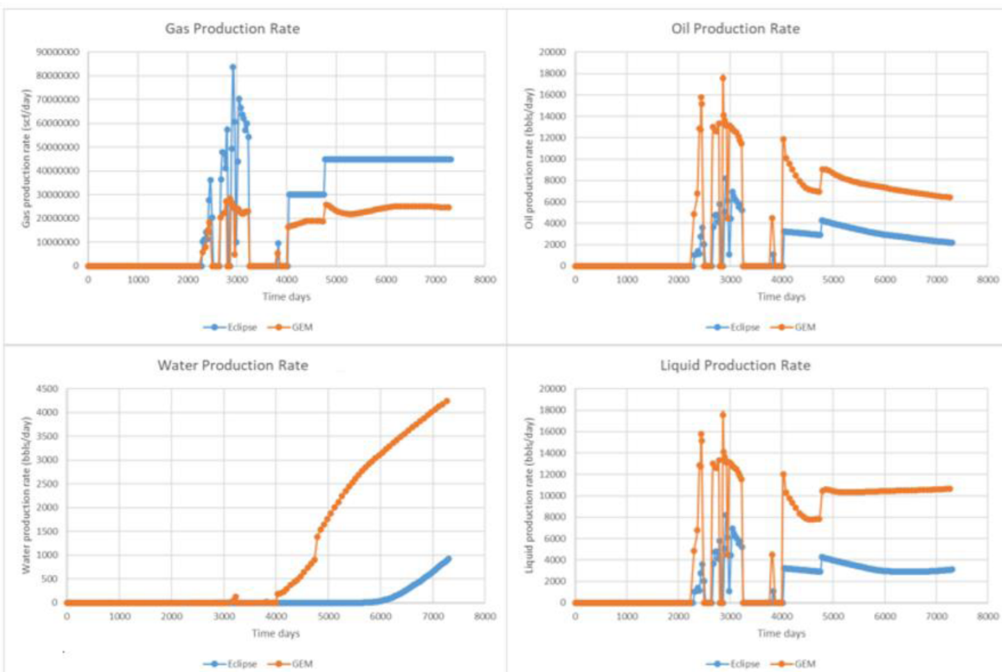

Figure 6—ECLIPSE and GEM Comparison – Well P6

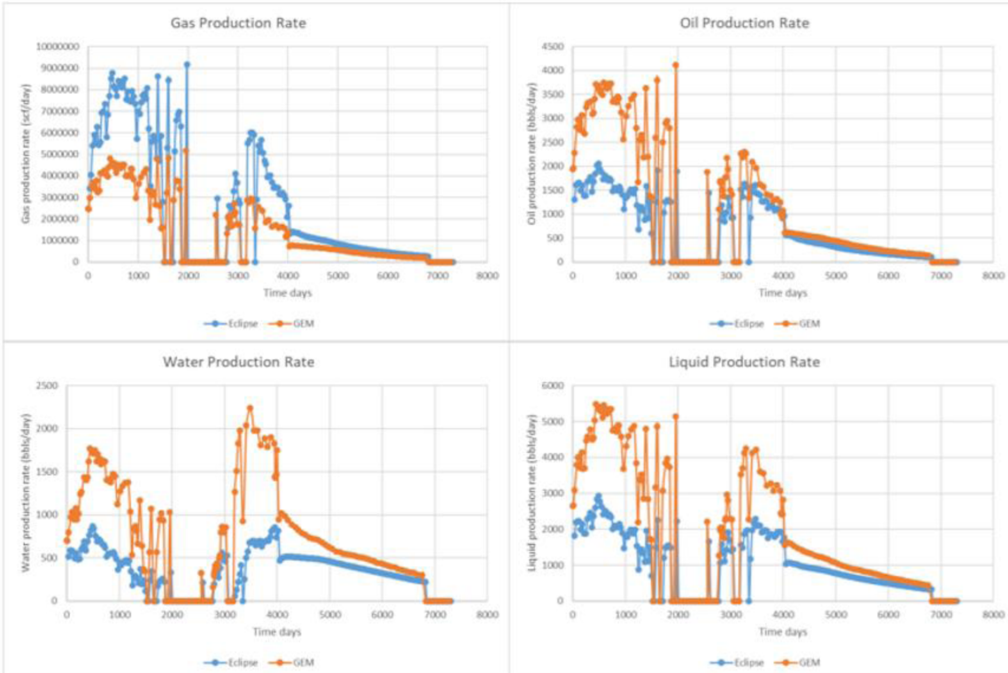

Figure 7—ECLIPSE and GEM Comparison – Well P7

Figures 5 to 7 show that the producers do not match closely in behavior between the ECLIPSE and GEM models. The GEM model tends to be pessimistic relative to the ECLIPSE model. Producer P6 has a much earlier water breakthrough timing and much higher evolution of the water production profile in the GEM model than in the ECLIPSE model. The resulting scaling risk evaluation using data from the GEM modelling will predict an earlier and higher scaling risk. Overall, the three producers undergo seawater breakthrough with a higher amount of water produced in the GEM model compared with the ECLIPSE model.

The reasons for the differences between the models may be attributed to the following:

1. The ECLIPSE model allows for input of different hydrocarbon compositions in different layers/zones (see Figure 4), whereas the GEM model can only use one set of hydrocarbon compositions for the whole reservoir. Thus, what was considered the most widely representative hydrocarbon composition was applied across the entirety of the GEM model, but this introduced local differences in hydrocarbon compositions between the models, which means differences in hydrocarbon phase densities and mobilities between the models.
2. Hydrocarbon compositions in the ECLIPSE model are not equilibrated at initial conditions with the aqueous and mineral phases, whereas in GEM the model should be initialized with all phases in equilibrium.

Despite the difference in the fluid phase properties in the two models identified above, the conversion of the geometric and petrophysical properties from ECLIPSE to GEM proved to be a success (as shown in Figures 8 and 9). Additionally, the initial saturation distributions were also successfully converted, as were the injection well flow rates, and despite the differences in some of the well production rates, the overall distribution map of saturations at the end of the simulation showed relatively minor differences (as shown in Figures 10 and 11). This shows that while there were discrepancies between the models in the flow rates in some of the producers, the contact between brine and rock was occurring in a similar fashion in the two models, and as we shall note below, it is this contact that is a key feature in determining the brine compositions in the producers.

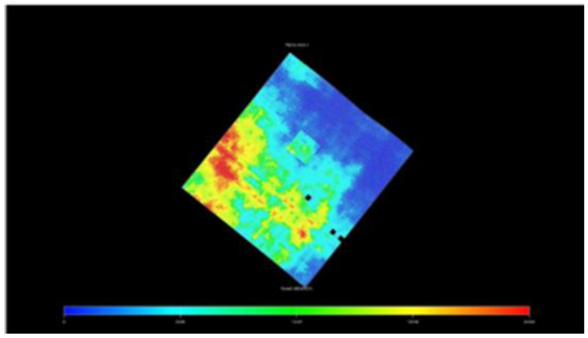

Figure 8—Horizontal permeability distribution in the ECLIPSE model layer 36

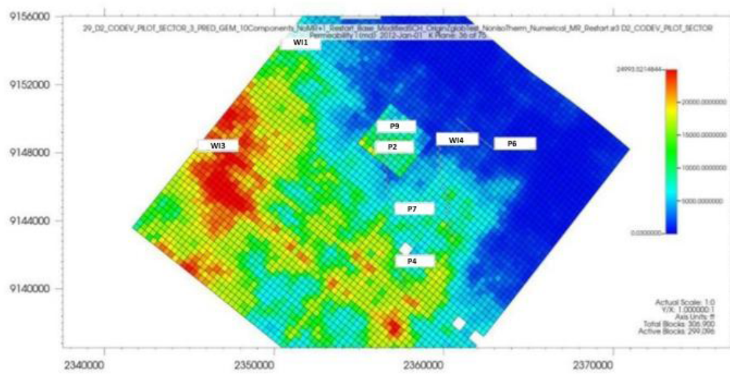

Figure 9—Horizontal permeability distribution in the GEM model layer 36

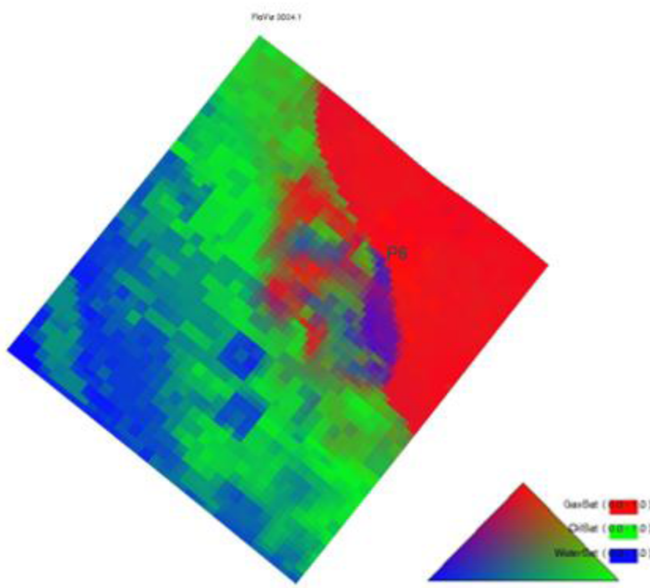

Figure 10—Ternary saturation distribution in the ECLIPSE model layer 57

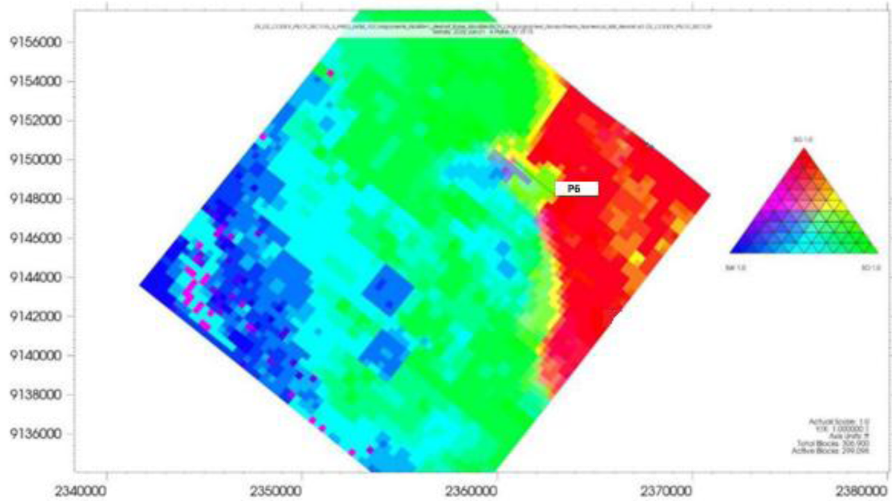

Figure 11—Ternary saturation distribution in the GEM model layer 57

Figures 12 to 14 show the comparisons between the ECLIPSE and GEM models of the seawater fractions in the three production wells that are calculated to cut seawater during the lifetime of the model. The simulations have been performed for wells P1, P6 and P7 because other wells do not undergo seawater breakthrough.

Producer P7 has the earliest seawater breakthrough, followed by P1 after around 1.5 years, and P6 has the latest seawater breakthrough - towards the very end of production according to the ECLIPSE model, although according to the GEM model the timing of seawater breakthrough for this well is similar to the timing for the other two wells. All producers in the GEM model produce more seawater than in the ECLIPSE model, and so the scaling risk prediction would be more conservative compared to the predictions with ECLIPSE (i.e. the output of the GEM modelling identifies a higher scaling risk based on seawater fraction than would the output of the ECLIPSE model for the same period). In Figures 12 to 14, All "0" values calculated during shut-in periods have been deleted for easier interpretation.

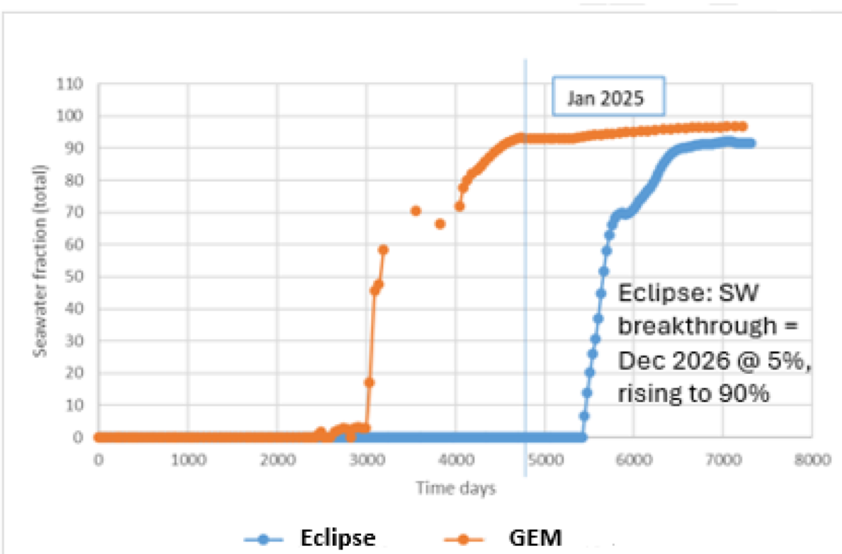

Figure 12—Seawater fraction for producer well P6

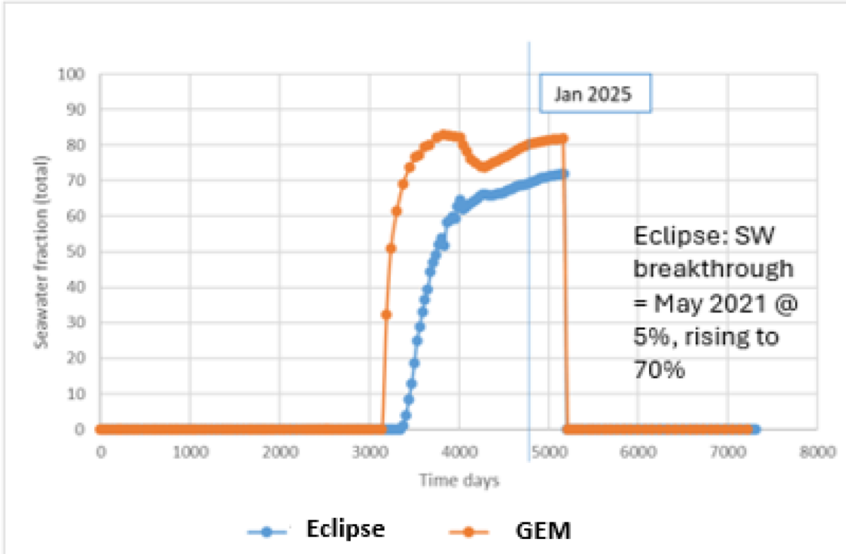

Figure 13—Seawater fraction for producer well P1

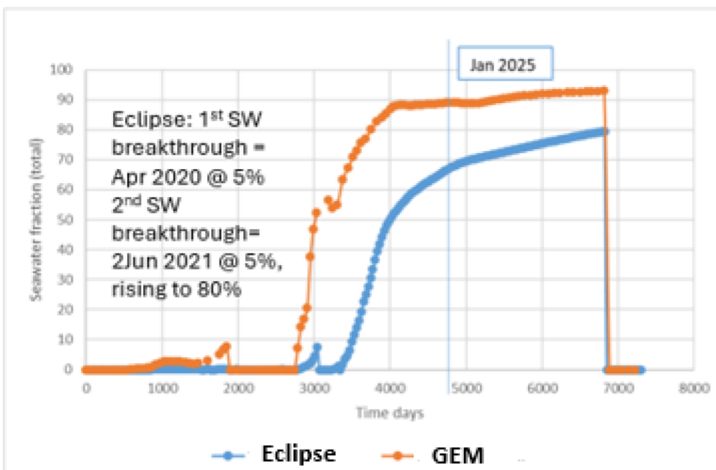

Figure 14—Seawater fraction for producer well P7

Ion Concentrations in produced water for three wells with seawater breakthrough – GEM Model

Figures 15 to 17 present the results of GEM modelling of full sulfate seawater injection. The key to understanding the scaling risk in production wells, in addition to the water flow rates (shown in Figures 12 to 14) and the injection water fractions (shown in Figures 12 to 14, January 2025 corresponds to day 4749 (13 years from start)) are the concentrations of all the ions that may be involved in mineral precipitation reactions, which not only depend on seawater fraction (and dilution with formation water), but also on any geochemical reactions that may have occurred as fluids transit the reservoir. These are shown for the three wells in question in Figures 15 to 17.

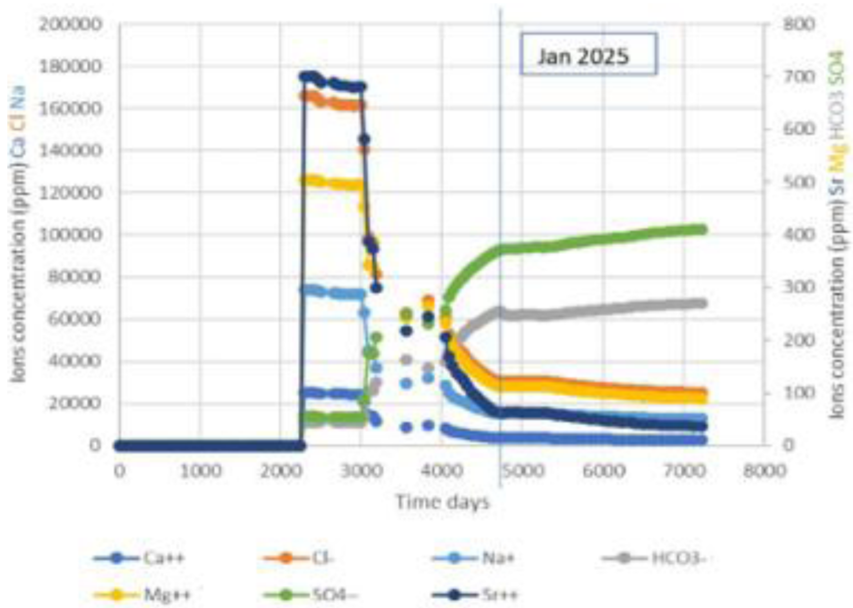

Figure 15—Ion concentrations in produced water in well P6

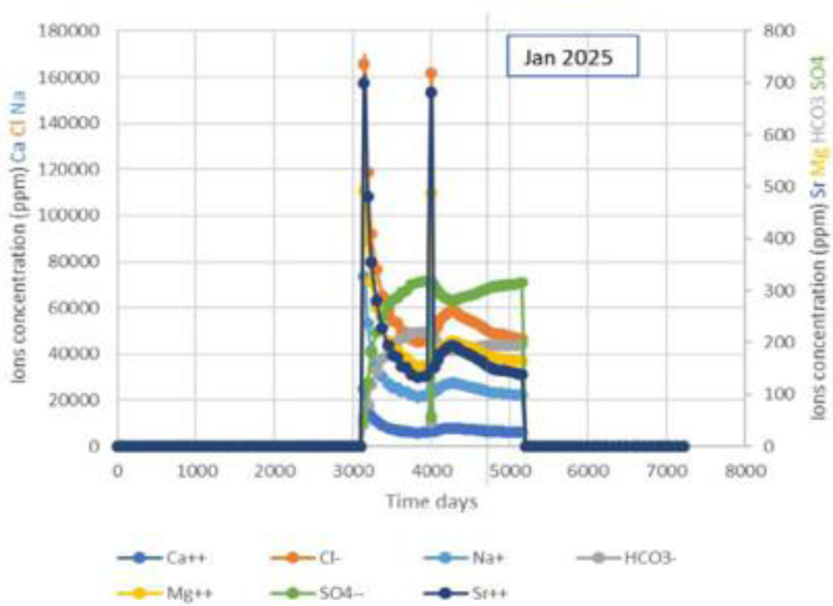

Figure 16—Ion concentrations in produced water in well P1

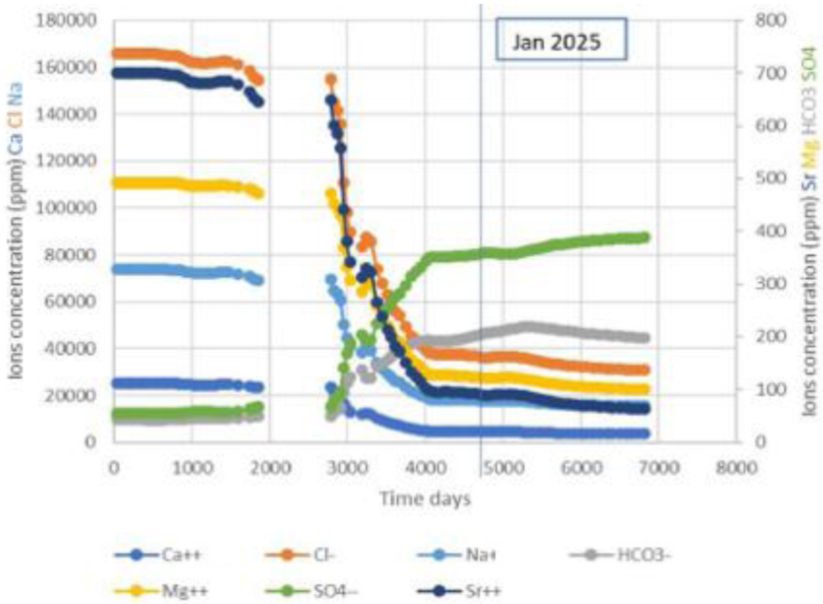

Figure 17—Ion concentrations in produced water in well P7

GEM model – Scaling risk prediction

The saturation ratio of $CaSO_4$ and $SrSO_4$ are calculated according to Eqs. 5 and 6, using the produced scaling ion concentrations, to measure the potential of their reaction, precipitating $CaSO_4$ or $SrSO_4$ in the production system.

$$SI = \log SR = \frac{Q}{Ksp} + \frac{[\mathrm{Ca}^{2+}][SO4^{2-}]}{Ksp} \qquad \text{Eq. 5}$$

$$SI = \log SR = \frac{Q}{Ksp} + \frac{[Sr^{2+}][SO4^{2-}]}{Ksp} \qquad \text{Eq. 6}$$

Where SI is the saturation index, SR is the saturation ratio, Q is the activity product and Ksp is the $CaSO_4$ and $SrSO_4$ equilibrium constants calculated using PHREEQC at the produced fluid flow conditions. Interpretation of the scaling risk based on saturation ratio (SR) calculation is as follows:

- SR = 1 equilibrium, no scaling risk
- SR > 1 precipitation, indicating a supersaturation scaling potential
- SR < 1 dissolution, no scaling risk

Produced ion concentrations can be used to calculate whether or not the produced brines will be oversaturated with respect to any scales at any point in the lifetime of the wells, and if so, where, when and how severe would that scaling risk be. A typical calculation is that of the Saturation Ratio (SR) for each mineral, the calculated value being the product of the ion activities divided by the mineral's equilibrium constant at well conditions (primarily temperature). In this work we calculate SR over time for each well for a single (downhole) temperature value, and assuming unit activity coefficients. SRs are calculated for anhydrite (Figures 18 to 20) and celestite (Figures 24 to 26) over the water production lifetime of the three wells for which ion concentrations are shown in Figures 15 to 17.

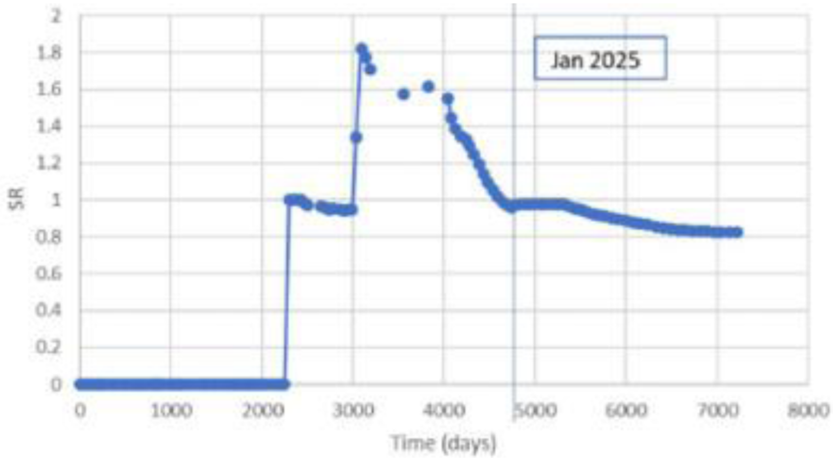

Figure 18—Saturation Ratio for $CaSO_4$ (anhydrite) in producer P6

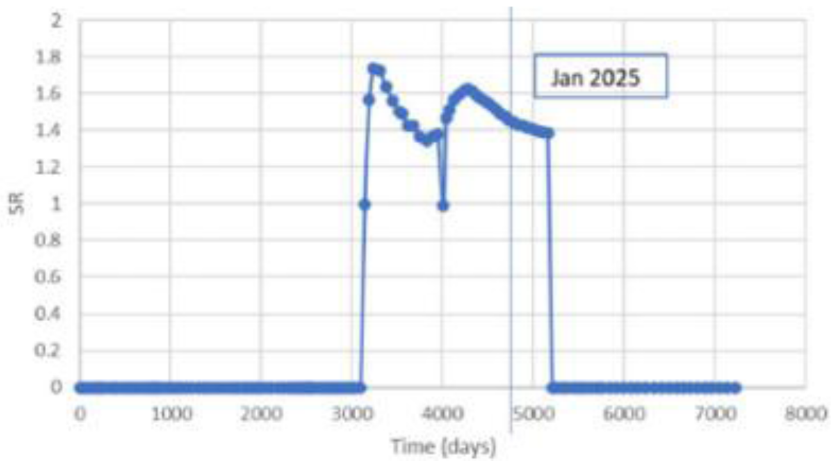

Figure 19—Saturation Ratio for $CaSO_4$ (anhydrite) in producer P1

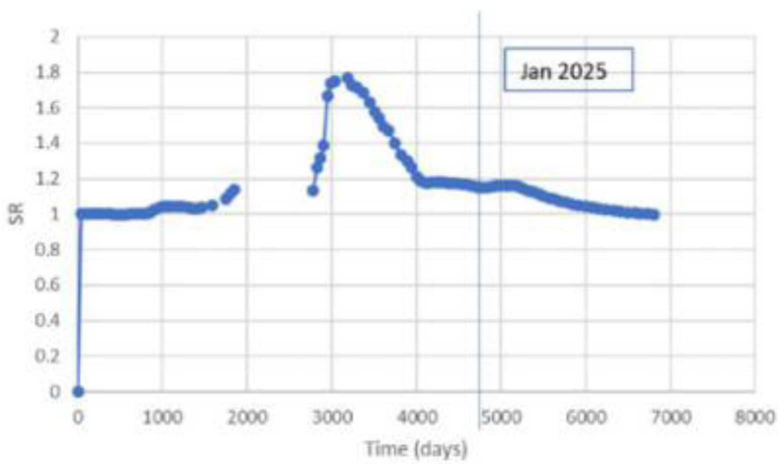

Figure 19a—Saturation Ratio for $CaSO_4$ (anhydrite) in producer P7

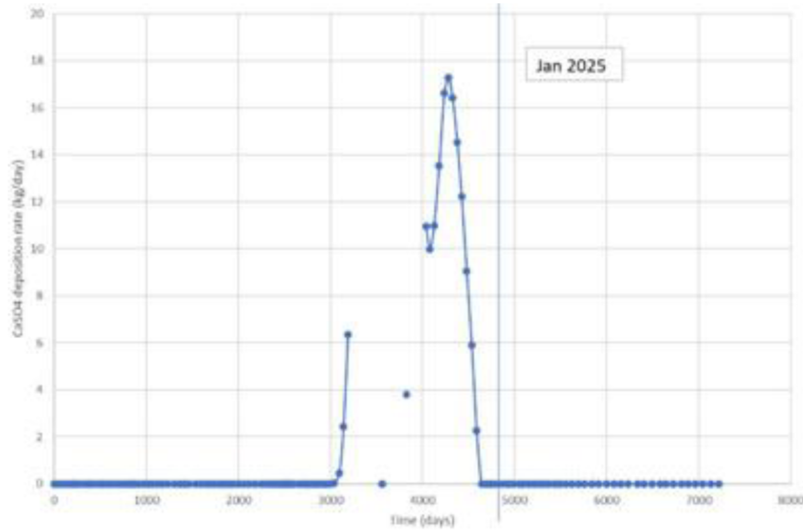

Figure 20—Deposition rate in kg/day for $CaSO_4$ (anhydrite) in producer P6

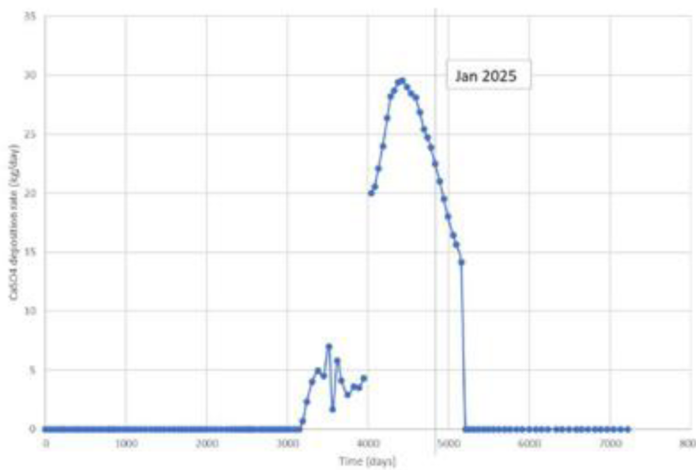

Figure 21—Deposition rate in kg/day for $CaSO_4$ (anhydrite) in producer P1

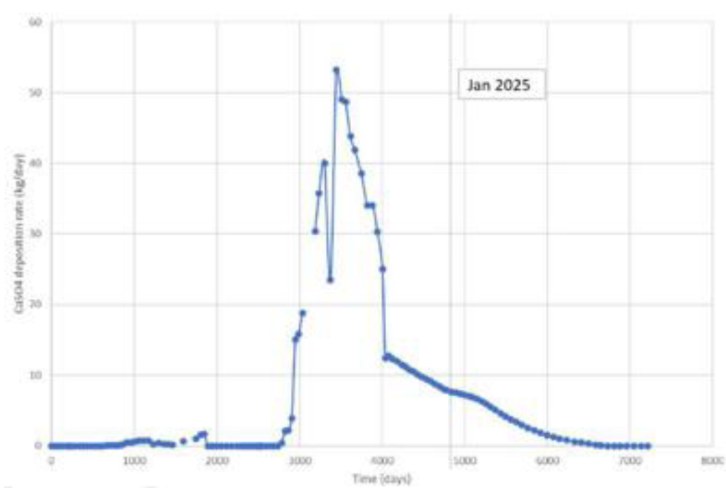

Figure 22—Deposition rate in kg/day for $CaSO_4$ (anhydrite) in producer P7

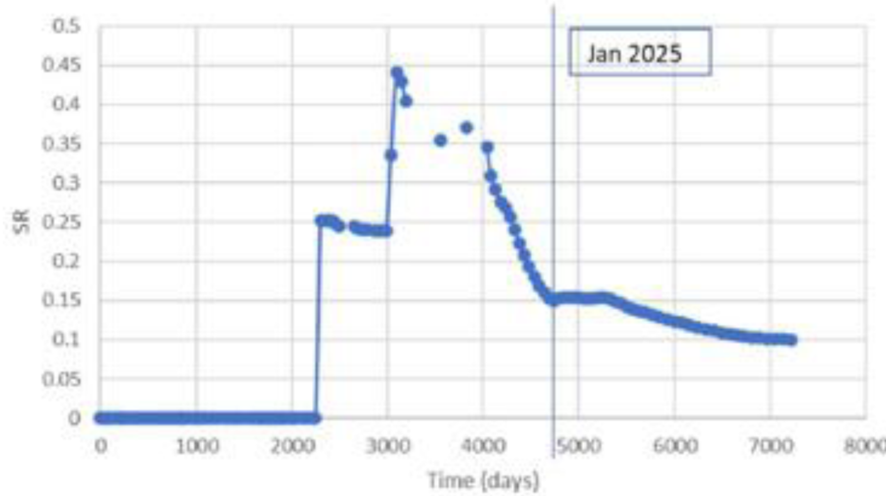

Figure 23—Saturation Ratio for $SrSO_4$ (celestite) in producer P6

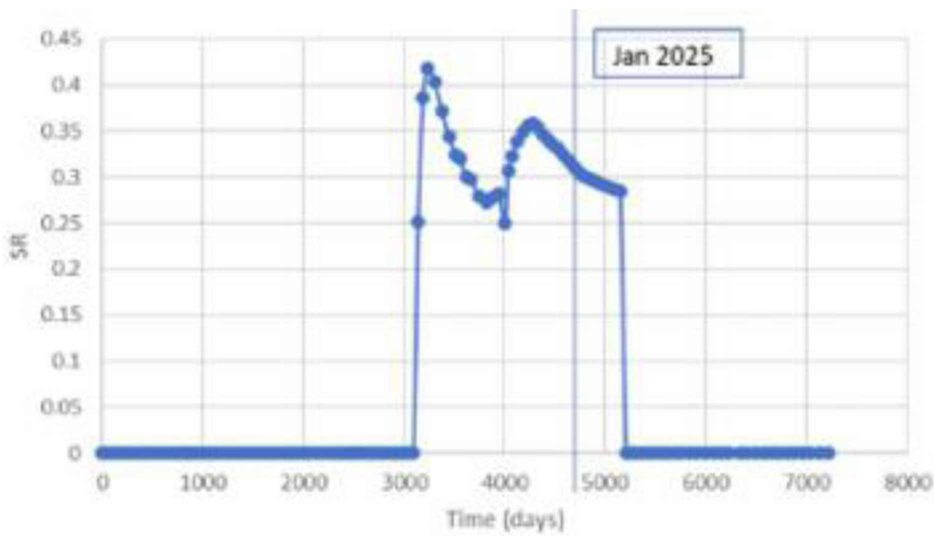

Figure 24—Saturation Ratio for $SrSO_4$ (celestite) in producer P1

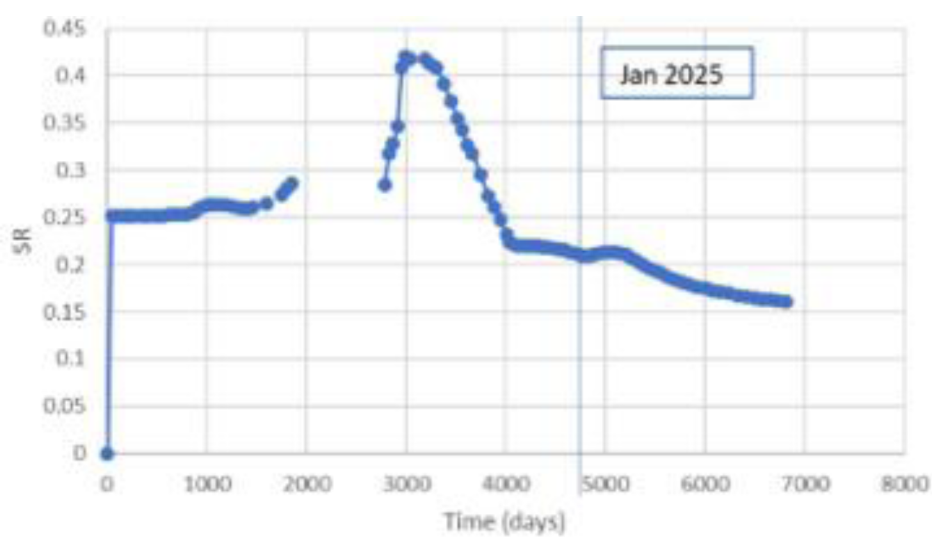

Figure 25—Saturation Ratio for $SrSO_4$ (celestite) in producer P7

For periods where SR=1, the brine is in equilibrium with the mineral: this is the case during initial formation water production in each case, and is as would be expected, since the formation brine will be equilibrated with the rock. When SR<1, if the mineral is already present then it will dissolve – as happens after seawater breakthrough in each case. Should SR>1, then precipitation of that mineral is to be expected: however, this does not occur in any well at any time. It may therefore be concluded from the profiles in Figures 24 to 26 that there is no scaling risk for celestite. That this is the case may be attributed to the combination of the injection water composition, and that geochemical reactions are predicted to occur in the reservoir to the extent that they alter the compositions of the brines ahead of production that once the brines flow into the wells they no longer have a scaling tendency. As for anhydrite scaling risk, all three wells may have $CaSO_4$ precipitation, the peak value of calculated SR for all three wells is 1.8 and Producer P1 stays above 1 for the period of interest; by contrast, Producers P6 and P7 experience a decrease in SR towards the end of production, with both wells observed to drop below SR=1 at the later stages of production.

Based on the SRs calculated and water production rates, deposition rates for the respective minerals can then be calculated to estimate the extent of scaling risk. The deposition rate of anhydrite is calculated and shown in Figures 20 to 22. It can be seen that for Producer P6 anhydrite starts to precipitate after year 8 (2020) of production; it then rises to the peak around year 11.5 (17.5 kg/day in October 2023 (day 4295)), and then it declines towards the end of production. For Producer P1, in January 2025 the deposition rate could be around 22 kg/day, which needs to be monitored for possible blockage due to the moderate scaling risk. As for Producer P7, it has a similar behavior to P6; however, the peak deposition rate is much higher (53 kg/day in April 2021) and the predicted deposition rate in January 2025 is estimated to be around 8 kg/day.

SRs are calculated for celestite $SrSO_4$ (Figures 23 to 25) over the water production lifetime of the three wells for which ion concentrations are shown in Figures 15 to 17.

According to GEM model, 3 producers experience seawater breakthrough (timings earlier). At the beginning of production, calcite dissolution and dolomite precipitation occur, leading to an increase of Ca and decrease of Mg – also stripping out SO_4 due to $CaSO_4$ precipitation afterwards (based on the validation of initial formation water equilibration from 1D GEM). No $SrSO_4$ scaling risk in any producers. From Jan 2025 - $CaSO_4$ has moderate deposition for P005 and P278 (22 kg/day and 8 kg/day, respectively); no $CaSO_4$ deposition for GP220.

Conclusions

- The results of this study are based solely on the provided sector model.
- Only matrix transport has been considered. Should there be "short circuit" high conductivity pathways, say through a fracture corridor, then the contact between injected brine and rock matrix could be significantly reduced, and the "natural desulfation" process might not occur.
- Based on the supplied water composition, during initial equilibration between the primary minerals and the formation water, the model required calcite dissolution and dolomite precipitation to occur; this led to an increase of Ca^{2+} concentration and a decrease of Mg^{2+} in the equilibrated brine – also stripping out SO_4^{2-} due to the equilibration with anhydrite.
- The model predicts that there will be no $SrSO_4$ scaling risk for any of the producers. However, there is $CaSO_4$ scaling risk for all three producers that cut seawater.
- Although all producers in the GEM model are predicted to have more water/seawater produced compared to the ECLIPSE.
- GEM model, as of January 2025,
    - Producer P6 has no precipitation
    - Producer P1 and P7 have $CaSO_4$ deposition rate at 22 kg/day and 8 kg/day respectively.

- Recommendation to perform the following monitoring:
- Seawater breakthrough and its for wells P2, P3, P4, P5, P8 and P9; ions to be monitored: SO_4, Ca, Sr, Na, Cl
  - Validation of the results of the GEM model
  - Evolution of the $CaSO_4$ scaling with time at the producers
  - Verification of the occurrence of the $SrSO_4$ scaling at the producers
- Based on the geochemical modelling results (time of scale occurrence, duration of the scaling risk and amount of scale deposits) have shown that injection of desulfated seawater is not necessary especially because of the high CAPEX of the SRU Unit.
- Options of $CaSO_4$ mitigation:
  - Scale dissolver
  - Acid wash
  - Scale inhibitor squeeze treatments might be the complementary scale mitigation that could be applied.

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